

Yuting Chen

Assistant Professor of Statistics

Department of Mathematics and Statistics, Eastern Kentucky University

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Research Interests

Small area estimation; survey statistics and data integration; mixed-effects models; uncertainty quantification; poverty mapping.

Academic Appointments

Assistant Professor of Statistics

August 2025 - Present

Department of Mathematics and Statistics, Eastern Kentucky University

Education

Ph.D. in Survey and Data Science

2020 - 2025

University of Maryland, College Park

Dissertation: Robust and Flexible Methods for Small Area Estimation

Advisor: Partha Lahiri

B.S. in Statistics

2016 - 2020

Jiangxi University of Finance and Economics

Publications

Peer-Reviewed Journal Articles

1. **Chen, Y.**, Hirose, M. Y., & Lahiri, P. (2026). Impact of existence and nonexistence of pivot on the coverage of empirical best linear prediction intervals for small areas. *Journal of the American Statistical Association*, advance online publication. [DOI](#)
2. **Chen, Y.**, Lahiri, P., & Salvati, N. (2025). Effects of model misspecification on small area estimators. *Survey Methodology*, 51(2), 647-657. [PDF](#)
3. Liu, X., Liu, Y., Laketa, P., Nagy, S., & **Chen, Y.** (2025). Exact and approximate computation of the scatter halfspace depth. *Computational Statistics*, 40, 547-572. [DOI](#)
4. Li, H., Liu, X., **Chen, Y.**, & Fan, Y. (2022). Testing for serial correlation in autoregressive exogenous models with possible GARCH errors. *Entropy*, 24(8), 1076. [DOI](#)
5. Yuan, Y., Liu, X., **Chen, Y.**, Dong, Y., & Liu, Y. (2022). On similarity of the sample projection depth contours and its application. *Communications in Statistics - Theory and Methods*, 51(7), 1919-1935. [DOI](#)

Professional Article

6. **Chen, Y.**, & Li, H. (2026). Variance component estimation under a general area-level model. *The Survey Statistician*, 93, 52-61. [PDF](#)

Accepted for Publication

Chen, Y., Lahiri, P., & Salvati, N. (accepted for publication). Empirical best prediction of poverty indicators via nested error regression with high-dimensional parameters. *Journal of the Royal Statistical Society: Series A*. [arXiv:2502.15933](https://arxiv.org/abs/2502.15933)

Invited Talks & Conference Presentations

Invited and Keynote Talks

Invited Speaker

2026

"Empirical Best Prediction of Poverty Indicators via Nested Error Regression with High Dimensional Parameters" - Key Laboratory of Financial Data Science, Jiangxi University of Finance and Economics, July 2

Instructor

2026

"Practical SAE: Implementing Solutions Using Multiple Data Sources" - IASS-ISEC International Workshop, hybrid, May 18-21

Keynote Speaker

2026

"Estimation of Poverty in Small Areas" - Mu Sigma Rho Initiation, Eastern Kentucky University, Richmond, KY

Invited Speaker 2026

"The Coverage of Empirical Best Linear Prediction Intervals for Small Areas" - Biostatistics Seminar, University of Louisville, Louisville, KY

Invited Speaker 2024

"Empirical Best Method with Data-Driven Model Parameter Estimation for Non-Exchangeable Nested Error Regression Models" - Small Area Estimation Conference, Lima, Peru

Contributed Presentations and Posters

Contributed Speaker 2025

"Impact of the Existence and Nonexistence of Pivot on the Coverage of Empirical Best Linear Prediction Intervals for Small Areas" - Joint Statistical Meetings, Nashville, TN

Poster Presenter 2023

"Evaluation of Small-Area Estimators and Associated MSPE Estimators under Model Misspecification" - International Indian Statistical Association Conference, Golden, CO

Research & Consulting Experience

Short-Term Consultant 2024

The World Bank Group, Washington, DC

Research Intern 2023

Gallup, Omaha, NE

Honors, Fellowships & Awards

DC-AAPOR Student Paper Award 2025

ASA GSS/SRMS/SSS Student Paper Competition Award 2025

Outstanding Graduate Student Award, Washington Statistical Society 2024

Summer Research Fellowship, University of Maryland 2024 (declined)

Jacob K. Goldhaber Travel Grant, University of Maryland 2024

International Conference Student Support Award, University of Maryland 2024

IISA Travel Support, Colorado School of Mines 2023

Dean Fellowship, University of Maryland 2020

Teaching Experience

Instructor of Record - Eastern Kentucky University

R and Introductory Data Mining 2026

Applied Regression Analysis 2025 - 2026

Applied Statistics 2025 - 2026

Teaching Assistant - University of Maryland, College Park

An Introduction to Small Area Estimation Methods 2024

JPSM-World Bank Training Program in Sampling Techniques 2022

Statistics for Information Science 2021 - 2025

Professional Service

Journal Referee

Journal of the American Statistical Association; Statistics in Medicine; Calcutta Statistical Association Bulletin

Technical Skills & Memberships

Programming: R, SAS, Python

Tools: LaTeX, Git, Shiny, Qualtrics

Memberships: American Statistical Association; American Association for Public Opinion Research; International Chinese Statistical Association